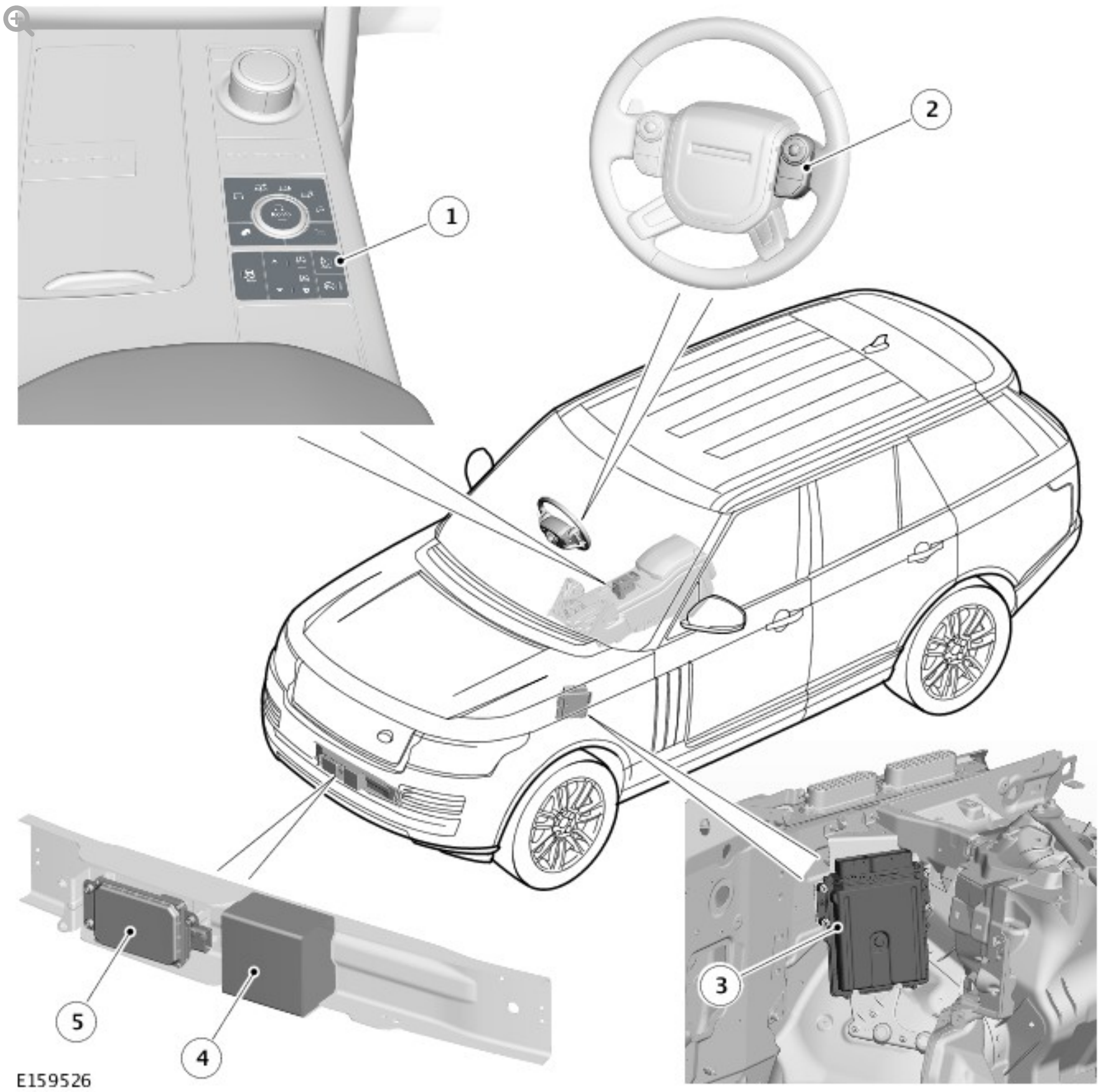
SPEED CONTROL – TDV6 3.0L DIESEL

DESCRIPTION AND OPERATION

COMPONENT LOCATION



ITEM	DESCRIPTION
1	Automatic speed limiter switch
2	Right steering wheel switchpack
3	Engine control module
4	Foam block
5	Adaptive speed control module

OVERVIEW

There are two variants of speed control available; a standard system and an adaptive system. The standard system maintains a set speed selected by the driver until operation is suspended or cancelled by a further input from the driver. The adaptive system includes the same functionality as the standard system, and has the following additional functions:

- Follow Mode: Reduces vehicle speed, to less than the set speed, in order to maintain a selected time gap behind a slower moving vehicle, then resumes the set speed once the way ahead is clear
- Queue Assist: Stops the vehicle a set distance behind the vehicle in front if it comes to a halt. When the vehicle in front moves away, a brief press of the accelerator pedal resumes adaptive speed control operation
- Forward Alert: Alerts the driver when rapidly approaching a slower moving vehicle
- Advanced Emergency Brake Assist: Partially applies the brakes in preparation for a sudden braking action, to improve brake response
- Intelligent Emergency Braking: Applies the brakes to reduce the impact velocity if a collision is unavoidable.

The standard speed control system is controlled by the ECM (engine control module) and switches on the right steering wheel switchpack. The adaptive speed control system is controlled by the ECM, the switches on the right steering wheel switchpack and the adaptive speed control module. Both systems also use the:

- Brake pedal switch. For additional information, refer to: (206-11 Brake Controls)
  Braking Control System (Description and Operation),
  Braking Control System - SDV6 3.0L Diesel - Hybrid Electric Vehicle (Description and Operation).
- APP (accelerator pedal position) sensor. For additional information, refer to:
  Electronic Engine Controls (303-14A Electronic Engine Controls - TDV6 3.0L Diesel, Description and Operation),
  Electronic Engine Controls (303-14F Electronic Engine Controls - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Description and Operation).
- ABS (anti-lock brake system) control module. For additional information, refer to: (206-11 Brake Controls)
  Braking Control System (Description and Operation),
  Braking Control System - SDV6 3.0L Diesel - Hybrid Electric Vehicle (Description and Operation).
- Transmission control switch. For additional information, refer to:
  External Controls (307-05A Automatic Transmission/Transaxle External Controls - Vehicles With: 8HP45 8-Speed Automatic Transmission AWD, Description and Operation),
  External Controls (307-05B Automatic Transmission/Transaxle External Controls - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Description and Operation).
- TCM (transmission control module). For additional information, refer to:
  Transmission Description (307-01A Automatic Transmission/Transaxle - Vehicles With: 8HP45 8-Speed Automatic Transmission AWD, Description and Operation),
  Transmission Description (307-01B Automatic Transmission/Transaxle - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Description and Operation).

- Terrain Response® switchpack.
  For additional information, refer to: Ride and Handling Optimization (204-06 Ride and Handling Optimization, Description and Operation).
- Instrument cluster. For additional information, refer to: (413-01 Instrument Cluster)
  Instrument Cluster (Description and Operation),
  Instrument Cluster (Description and Operation).

During speed control operation, the ECM controls vehicle speed by adjusting the engine throttle and fuel injection. The adaptive speed control system also uses the adaptive speed control module to scan the road ahead and, if necessary, signal the ABS control module to apply the brakes to slow the vehicle.

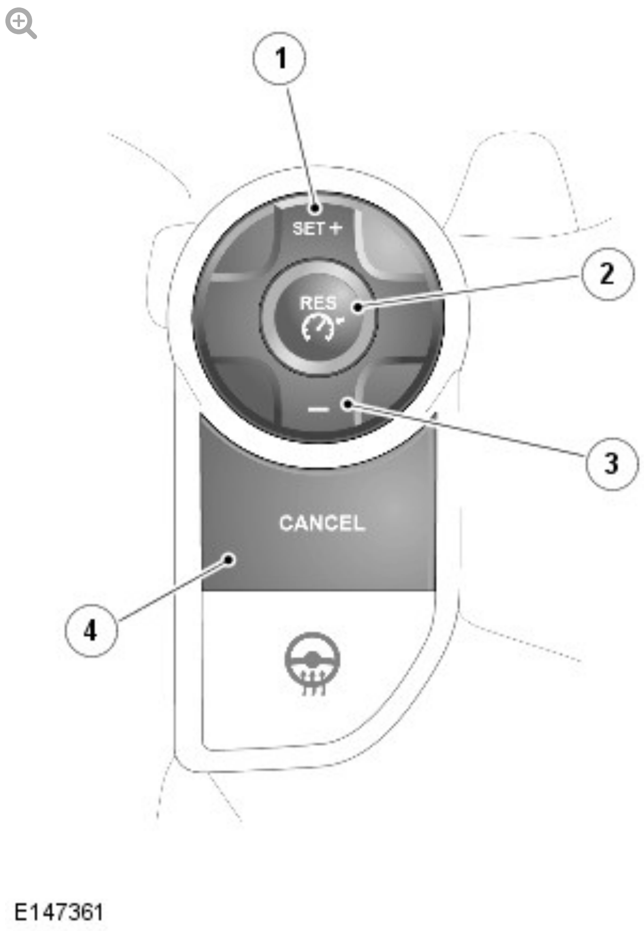
All vehicles also incorporate an ASL (automatic speed limiter) system. When a speed has been set, the engine will respond normally up to the set speed. Further accelerator pressure will not increase the speed beyond the set speed unless sudden, rapid acceleration (kickdown) is applied. The ASL system is controlled by the ECM, the speed control switches on the right steering wheel switchpack and an ASL switch.

The ASL system and the speed control system cannot be used simultaneously. The system in use when the ignition is turned off is automatically enabled when the ignition is turned back on. However the speed settings are not remembered and will have to be set again.

DESCRIPTION

STANDARD SPEED CONTROL

SPEED CONTROL SWITCHES

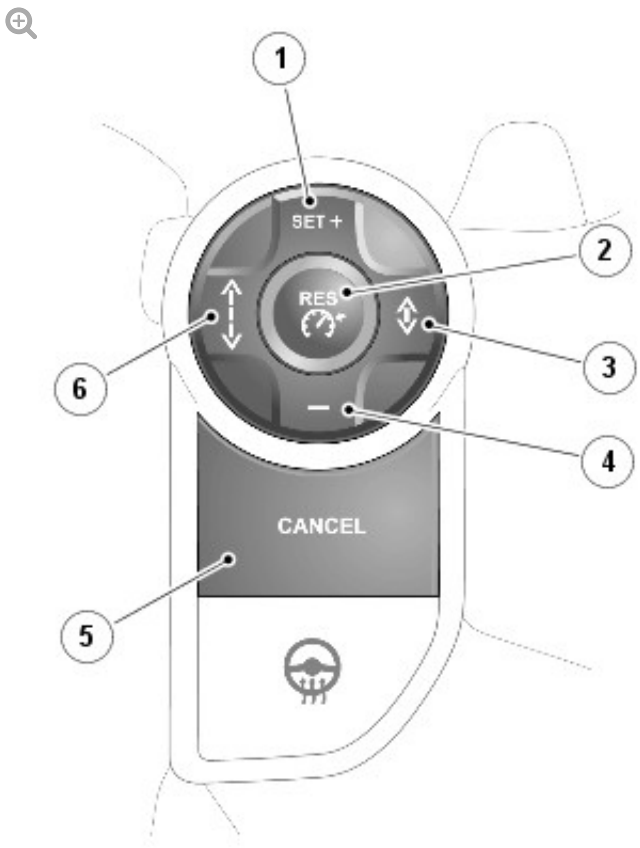


ITEM	DESCRIPTION
1	Set + (engage/accelerate) switch
2	RES (resume) switch
3	- (decelerate) switch
4	CANCEL switch

The standard speed control switches are located on the right steering wheel switchpack. The switches are resistive ladder type switches which vary the resistance of a 5 volt signal sent to them. The resistive ladder is wired to a LIN (local interconnect network) node which transmits the data to the CJB (central junction box) via the clockspring. The CJB then transmits the data on the high speed CAN (controller area network) powertrain systems.

ADAPTIVE SPEED CONTROL

SPEED CONTROL SWITCHES



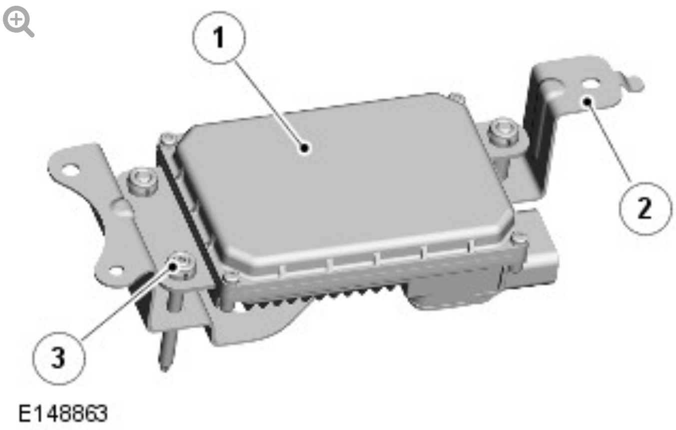
E147362

ITEM	DESCRIPTION
1	Set + (engage/accelerate) switch
2	RES (resume) switch
3	Time gap decrease switch
4	- (decelerate) switch
5	CANCEL switch
6	Time gap increase switch

The adaptive speed control switches are the same as used for standard speed control, with the addition of two time gap switches. The time gap switches allow the driver to adjust the follow mode and forward alert functions to one of four pre-set time gaps (gap 1 = 1 second; gap 2 = 1.4 seconds; gap 3 = 1.8 seconds; gap 4 = 2.2 seconds). The selected time gap is displayed in the message center when the time gap switches are operated.

When adaptive speed control is engaged, the time gap switches adjust the follow mode time gap. When adaptive speed control is disengaged, the time gap switches adjust the forward alert time gap (provided forward alert is enabled in the instrument cluster menu). The follow mode function defaults to the gap 3 setting each time the ignition is switched on. The forward alert function recalls the last selected time gap when the ignition is switched on.

ADAPTIVE SPEED CONTROL MODULE



ITEM	DESCRIPTION
1	Adaptive speed control module
2	Mounting bracket
3	Adjuster screw

The adaptive speed control module is attached to a bracket located in the armature of the front bumper, to the right of the vehicle centerline. A foam block in the bumper armature eliminates radar signal reflections within the armature.

The adaptive speed control module contains a forward looking radar transceiver together with related controlling hardware and software. A high speed CAN powertrain systems connection allows the adaptive speed control module to communicate with other system control modules. Power supplies to the adaptive speed control module are from the extended ignition relay in the RJB (rear junction box) and the ignition relay in the EJB (engine junction box).

The adaptive speed control module is active whenever the ignition is on, even if speed control is not engaged, in order to operate the forward alert, advanced emergency brake assist and intelligent emergency braking functions. With the ignition on, the adaptive speed control module is powered up but no radar transmissions are emitted until the vehicle is in motion.

The adaptive speed control module transmits a radar beam forward of the vehicle and detects the returning signals reflected off other vehicles and objects ahead. The radar beam is electronically scanned at a rate of 20 sweeps/second across a total arc of 20° centered on the longitudinal axis of the vehicle. The radar operates at millimetric wavelengths (76 - 77 GHz) and transmits a frequency modulated continuous wave signal at a relatively low power level (no high power pulses). The control module detects the range, relative velocity and angle of objects within the scanned arc for up to a distance of approximately 130 meters (426.5 feet).

The adaptive speed control module compares vehicle speed data from the ABS control module with the relative speed of an external object as detected by the radar to ascertain whether the object is stationary or not. If tires are fitted which are different in diameter from those specified for the vehicle, the vehicle speed, calculated by the ABS module, will not be the true road speed. This situation may cause stationary objects to be falsely identified as moving vehicles and result in automatic deceleration on a clear road.

The adaptive speed control module continuously monitors both moving vehicles and stationary objects to determine if it can 'see' normally. If the control module can detect only a few objects, because it is physically blocked (for example by an accumulation of snow or mud on the lower grill of the front bumper, or an incorrectly located number plate), it is incorrectly aligned, or there are few roadside objects or other vehicles in the area, it may determine that it is blocked. The control module then inhibits adaptive speed control and records a DTC (diagnostic trouble code). The system will reset once the cause of the blockage is removed. In this case the control module is not faulty and should not be replaced.

If the adaptive speed control module is replaced in service it must be mechanically aligned vertically using the adjuster screw. Horizontal alignment is achieved by putting the control module into service mode using approved Land Rover diagnostic equipment. The vehicle then needs to be driven for a short period while the control module calibrates itself. For additional information, refer to: Speed Control Sensor Adjustment (310-03D Speed Control - TDV8 4.4L Diesel, General Procedures).

AUTOMATIC SPEED LIMITER

ASL SWITCH

The ASL switch is integrated into the Terrain Response® switchpack in the floor console. An amber LED (light emitting diode) in the ASL switch remains illuminated while the ASL is selected on. Selection of the ASL switch is transmitted from the Terrain Response® switchpack to the ECM on the high speed CAN chassis systems.

INSTRUMENT CLUSTER WARNING INDICATORS

SPEED CONTROL WARNING INDICATOR

E147363

The speed control warning indicator illuminates when standard or adaptive speed control is engaged.

FOLLOW MODE WARNING INDICATOR

E122343

The follow mode warning indicator illuminates when the adaptive speed control system is engaged and goes into follow mode. When the system is engaged, the time gap setting is displayed in the message center in the form of a vehicle with 1 to 4 bars in front of it, where the number of bars correspond to the current time gap setting.

FORWARD ALERT WARNING INDICATOR

E122344

The forward alert warning indicator illuminates if forward alert is enabled in the instrument cluster menu.

ASL WARNING INDICATOR

E147868

The speed control warning indicator illuminates when ASL is selected on and a speed limit is set in the message center.

OPERATION

For standard or adaptive speed control to engage, the vehicle must be in the following configuration:

- Transmission in D (drive)
- Transfer case in high range
- Terrain Response® in the general (special programs off) or grass/gravel/snow program
- HDC (hill descent control) off
- ASL off
- Traveling at 20 mph (32 km/h) or more.

STANDARD SPEED CONTROL

ENGAGE SPEED CONTROL

Speed control is engaged by pressing the SET **+** switch. On receipt of the switch input, the ECM adopts the current vehicle speed as the speed control set speed. When speed control is engaged the instrument cluster illuminates the speed control warning indicator and the set speed is displayed in the message center.

The set speed is cleared when the ignition is switched off or if the speed control system develops a fault.

SET SPEED ADJUSTMENT

Once engaged the set speed can be increased by one of 3 methods:

- Press and hold the SET **+** switch which will cause the vehicle to accelerate. When the switch is released the attained speed will be the new set speed.
- Repeatedly pressing the SET **+** switch. Each press increases the vehicle speed by 1 mph (2 km/h).
- Use the accelerator pedal to increase the vehicle speed. Once the

required speed is reached, a single press of the SET **+** switch will set the cruise speed.

The set speed can be reduced by pressing and holding the decelerate **-** switch until the required speed is reached. When the switch is released the speed is reset at that value. The set speed can be reduced incrementally by pressing and releasing the decelerate **-** switch. Each press will reduce the speed by 1 mph (2 km/h).

Speed control adjustment is limited around the current vehicle speed. This means speed adjustment won't occur when resuming or if the vehicle speed is noticeably different to set speed. This is noticeable when resuming speed control and a large difference exists between the set speed and the current speed. The driver can set a new speed, but can't adjust the old speed.

SUSPENDING SPEED CONTROL

Suspending speed control means speed control has been deactivated without loss of the set speed. Speed control is suspended when one of the following occurs:

- Vehicle speed falls below 20 mph (32 km/h)
- The brake pedal is pressed
- The CANCEL switch is pressed
- N (neutral), R (reverse) or P (park) is selected on the TCS (transmission control switch)
- The EPB (electronic parking brake) is applied
- If the accelerator pedal is used to accelerate beyond the set speed for too long (more than 5 minutes)
- Engine speed too high (cancel limit set just below red line)
- The ABS control module activates ETC (electronic traction control) or DSC (dynamic stability control)
- HDC is selected

- Terrain Response® adopts a special program other than grass/gravel/snow
- The transfer case is selected to low range.

RESUMING SPEED CONTROL

A single press of the RES switch will resume speed control at the memorized set speed.

ADAPTIVE SPEED CONTROL

WARNING:

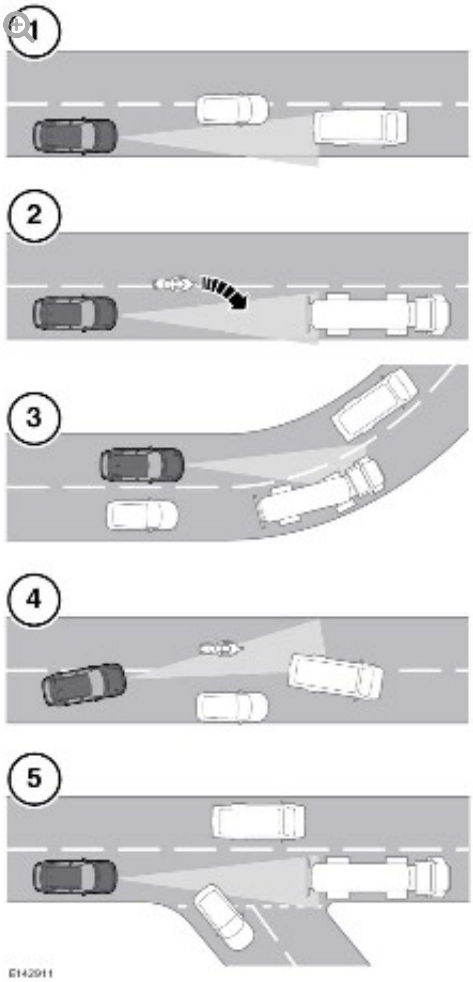
> Adaptive speed control system is not a collision warning or avoidance system. At all times the driver must be prepared to intervene to control the vehicle speed.

Adaptive speed control operation is similar to the standard speed control system for engaging, suspending and resuming speed control. The main differences between the two systems is that adaptive speed control remains engaged below 20 mph (32 km/h) and uses the brakes to slow the vehicle, to a halt if necessary.

If adaptive speed control is suspended by pressing the CANCEL switch or the brake pedal, it can be re-engaged by pressing the RES switch once vehicle speed is above 7 mph (11 km/h).

SYSTEM RESTRICTIONS

The adaptive speed control system is only intended to provide enhanced speed control in certain restricted conditions. The following illustration shows circumstances where the adaptive speed control system may brake late or unexpectedly. The driver is required to intervene in these situations.



1 Driving on a different line to the vehicle in front.

2 Vehicles that merge into the same lane are only detected once they have moved fully into the lane.

3 On bends in the road there can be issues with detection of the vehicle in front when going into and coming out of a bend.

4 Moving around a stationary vehicle may cause uncertainty as to which vehicle is being followed.

5 A vehicle in front moving out of your lane may cause uncertainty as to which vehicle is being followed.

The system will detect but not react to the following:

- Vehicles in the oncoming lane
- Stationary vehicles
- Pedestrians

- Vehicles not in the same lane
- Some smaller motorcycles.

In follow mode, some situations may cause target ambiguities for the detection system. These situations include:

- The nearby presence of a third vehicle when driving on a line slightly offset to the target vehicle
- Vehicles edging into the lane ahead which are not detected by the system until they have moved into the radar beam.

On the approach to, or exit from a bend, a target vehicle may be lost or a new target acquired as vehicles ahead change their angular position with respect to the adaptive speed control module. On a straight road, if the sensing vehicle is in follow mode below its selected set speed, losing the target vehicle will cause the sensing vehicle to accelerate to this set speed. This acceleration is undesirable either on, or entering a bend when the target is suddenly lost, and in this situation the system inhibits the resumption of the set speed.

FOLLOW MODE

A set speed is selected in the normal speed control manner and this speed is maintained until a slower vehicle is encountered in the lane ahead. When the vehicle ahead comes within the effective range of the radar, the system identifies it as a target vehicle. To indicate that the system is in follow mode, the follow mode warning indicator illuminates. When the distance between the two vehicles closes to the set time gap, the adaptive speed control system reduces engine speed and, if necessary, applies the brakes to maintain the set time gap.

Automatic braking is limited to approximately 30% of full pressure (0.3 G deceleration) and is intended to provide a smooth, gradual deceleration in follow mode conditions.

Follow mode is effectively a closed loop system. If several vehicles are ahead, the closest vehicle is chosen as the target to follow. If the target vehicle moves out of radar range, or if either vehicle changes lane, the system exits follow mode and extinguishes the follow mode warning indicator. The adaptive speed control system then increases vehicle speed back to the original set speed.

QUEUE ASSIST

When adaptive speed control is engaged, if the target vehicle comes to a halt the adaptive speed control module reduces engine speed and applies the brakes to halt the sensing vehicle and suspends speed control operation. The control module also applies the EPB if one of the following occurs:

- Adaptive speed control is disengaged with the CANCEL switch
- The vehicle is stationary for more than 2 minutes
- Driver exit is detected (assumed if the driver seatbelt is unbuckled and the driver door is opened)
- A system fault is detected.

On HEVs (hybrid electric vehicles), full use is made of regenerative braking during queue assist operation. Any additional braking effort required is produced by hydraulic application of the brakes.

When the target vehicle moves away, pressing the accelerator pedal for a minimum of 0.5 second resumes speed control operation. The sensing vehicle then follows the target vehicle in follow mode or until it stops again, or, if it can no longer detect a target, accelerates back up to the set speed.

NOTE:

> It may be necessary to press the accelerator pedal for longer to ensure a restart on HEVs.

FORWARD ALERT

If an object is detected close ahead, a warning tone will be emitted and a message will be displayed. The brakes will not be applied.

The forward alert system is active when the warning indicator in the instrument cluster is illuminated. Forward alert does not initiate any action, the driver must take the appropriate action when the message is displayed. The system monitors driver intervention and may not issue the warning if the driver has taken the appropriate action (braking, steering or indicating) early enough.

The sensitivity of the warning can be changed using the gap increase and decrease switches when the adaptive cruise control is not engaged. Confirmation of the gap change is given in the message center.

Forward alert can be switched on or off using the instrument cluster menu.

ADVANCED EMERGENCY BRAKE ASSIST

Advanced EBA (emergency brake assist) improves braking response during emergency braking, and is an enhancement of the electronic brake prefill and the EBA functions controlled by the ABS control module. For additional information, refer to: (206-11 Brake Controls)

Braking Control System (Description and Operation),
Braking Control System - SDV6 3.0L Diesel - Hybrid Electric Vehicle (Description and Operation).

With advanced EBA, if the risk of a collision increases after the vehicle passes the forward alert activation threshold, the adaptive speed control module signals the ABS control module to activate the electronic brake prefill and EBA functions. With advanced EBA, the electronic brake prefill function operates in two stages:

- Stage 1:
  - Applies a brake pressure of approximately 3 bar (43.5 lbf/in^2) to all of

the brakes

    - Lowers the EBA activation threshold in the ABS control module
    - Signals the CJB on the high speed CAN powertrain systems to activate the stop lamps

- Stage 2 increases the brake pressure to approximately 10 bar (145 lbf/in²).

Stage 1 is activated if the collision risk increases to a level considered as credible, with the accelerator pedal pressed. Stage 2 is activated if the collision risk increases further, to a level considered as imminent, with the accelerator pedal pressed. If the accelerator pedal has already been released when the collision risk increased to the credible level, the two stages of electronic brake prefill are activated simultaneously, to give an immediate brake pressure of 10 bar (145 lbf/in²).

While the second stage of electronic brake prefill is active, if the brake pedal is then pressed quickly, the ABS control module activates EBA, even if only light pressure is applied to the pedal.

Advanced EBA is available at vehicle speeds above 5 mph (7 km/h) and will function even when forward alert and adaptive cruise control are switched off.

If a fault occurs in the system a related message is displayed in the message center. Advanced EBA will not be available until the fault is rectified.

INTELLIGENT EMERGENCY BRAKING

Intelligent emergency braking reduces the impact velocity if the adaptive speed control module determines that a collision is unavoidable, even with driver intervention. It is not designed to prevent accidents from occurring. Typical activation of intelligent emergency braking is approximately 0.8 second before impact. During this 0.8 second period the speed of the sensing vehicle is reduced, thus reducing the impact energy.

If the adaptive speed control module determines a collision is unavoidable, it signals the ABS control module to apply emergency brake pressure to all of the brakes. On vehicles without active seatbelts, emergency brake pressure is 50 bar (725 lbf/in²) maximum. On vehicles with active seatbelts, emergency brake pressure is 100 bar (1450 lbf/in²) maximum.

When intelligent emergency braking is initiated the adaptive speed control module also signals the active seatbelt control modules to tighten the front seatbelts.
For additional information, refer to: Seatbelt System (501-20A Seatbelt System, Description and Operation).

After the intelligent emergency braking function has activated, it is then disabled, as the vehicle is assumed to be damaged. After the vehicle has been repaired, the intelligent emergency braking function can be re-enabled using Land Rover approved diagnostic equipment.

The intelligent emergency braking function operates at vehicle speeds down to zero, and will function even when forward alert and adaptive cruise control are switched off.

If the intelligent emergency braking feature is disabled for any reason, a related message is displayed in the message center.

AUTOMATIC SPEED LIMITER

With the ASL switch selected on, the ASL system can be operated in a similar manner to the cruise control system. A speed limit between 20 and 127 mph (32 and 204 km/h) can be set, with the engine running and the vehicle stationary or moving, using the speed control SET **+** and **-** switches (in ASL operation, the SET **+** and the **-** switches change the set speed limit in steps of 5 mph (8 km/h), instead of the 1 mph (2 km/h) steps of the cruise control system). The ASL warning indicator illuminates and the set speed limit is displayed in the message center.

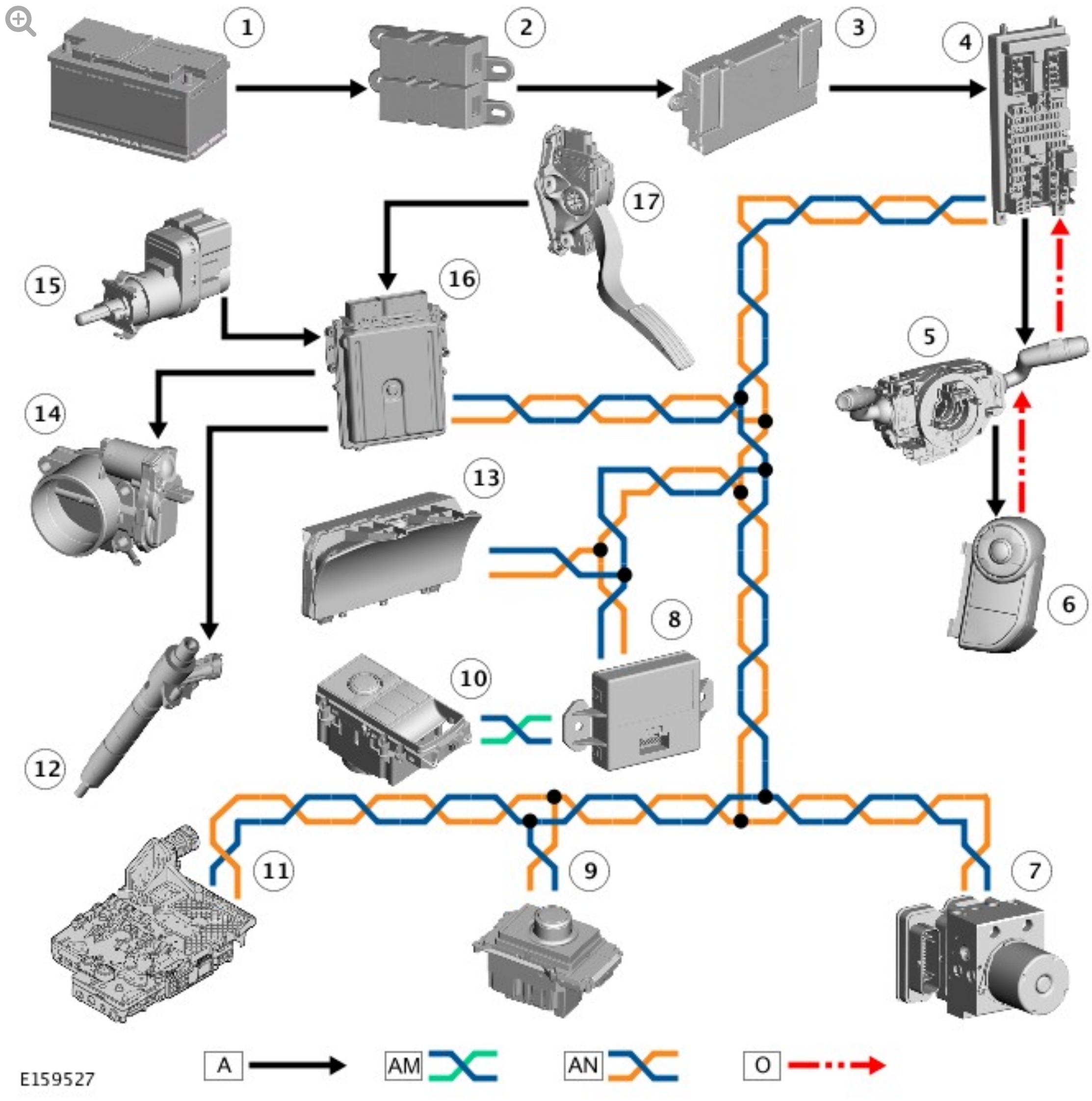
If the vehicle reaches the set speed limit, the ECM intervenes to govern vehicle speed at the set speed limit.

NOTE:

In certain conditions, such as a steep downhill gradient, the vehicle speed may exceed the set speed limit. This is because engine braking is unable to maintain or reduce the vehicle speed.

The ASL can be disengaged and re-engaged using the speed control CANCEL and RES switches respectively. ASL also disengages if kickdown is used on the accelerator pedal. Pressing the ASL switch cancels ASL operation and the speed limit will need to be reset the next time the switch is selected on.

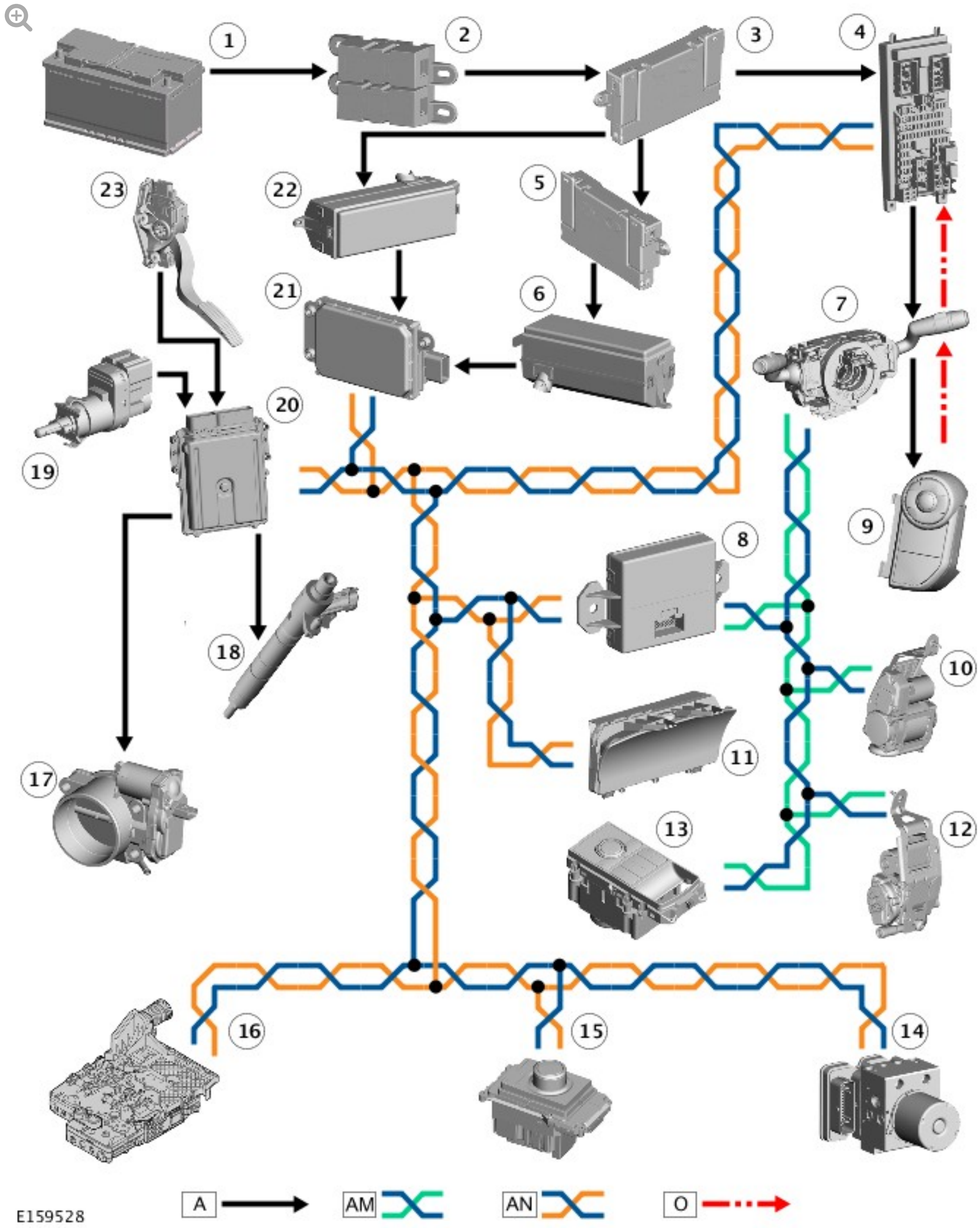
CONTROL DIAGRAM - STANDARD SPEED CONTROL



A = HARDWIRED; AM = HIGH SPEED CAN CHASSIS SYSTEMS; AN = HIGH SPEED CAN POWERTRAIN SYSTEMS; O = LIN BUS.

ITEM	DESCRIPTION
1	Battery
2	Battery junction box 2
3	Battery junction box
4	Central junction box
5	Clockspring
6	Right steering wheel switchpack
7	Anti-lock brake system control module
8	Gateway module
9	Transmission control switch
10	Terrain Response® switchpack
11	Transmission control module
12	Fuel injectors
13	Instrument cluster
14	Electric throttle
15	Brake pedal switch
16	Engine control module
17	Accelerator pedal position sensor

CONTROL DIAGRAM – ADAPTIVE SPEED CONTROL



A = HARDWIRED; AM = HIGH SPEED CAN CHASSIS SYSTEMS; AN = HIGH SPEED CAN POWERTRAIN SYSTEMS; O = LIN BUS.

ITEM	DESCRIPTION
1	Battery
2	Battery junction box 2
3	Battery junction box
4	Central junction box

5	Auxiliary junction box
6	Engine junction box
7	Clockspring and steering angle sensor
8	Gateway module
9	Right steering wheel switchpack
10	Left active seatbelt retractor
11	Instrument cluster
12	Right active seatbelt retractor
13	Terrain Response® switchpack
14	Anti-lock brake system control module
15	Transmission control switch
16	Transmission control module
17	Electric throttle
18	Fuel injectors
19	Brake pedal switch
20	Engine control module
21	Adaptive speed control module
22	Rear junction box
23	Accelerator pedal position sensor